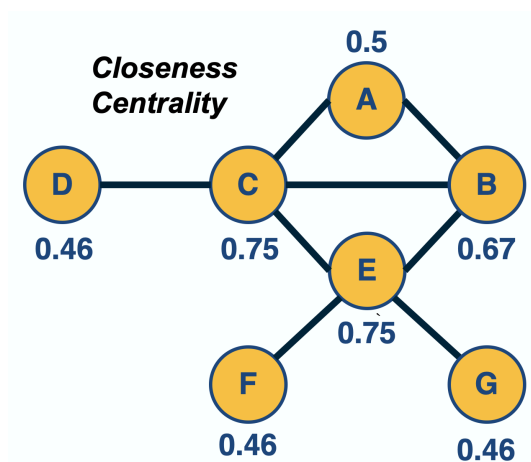


L6: Closeness Centrality and Harmonic Centrality

Closeness Centrality



The previous centrality metrics are all based on the direct connections of a node. Another group of centrality metrics focuses on network paths. Paths represent "routes" over which something is transferred through a network (e.g., information, pathogens, materials). Consequently, another way to think about the centrality of a node is based on how good the routes are of that node to the rest of the network (or how many are the routes that traverse a node).

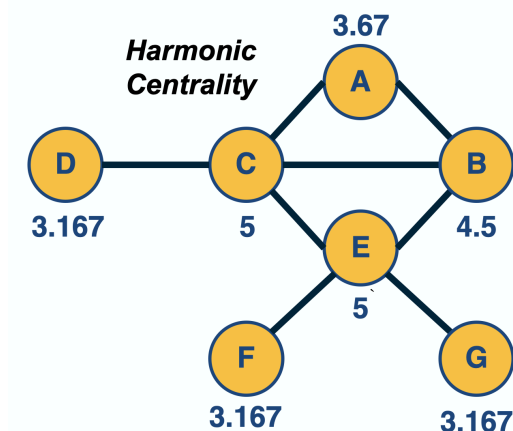
A commonly used path-based metric is closeness centrality. It is based on the length (number of hops) of the shortest-path between a node i and a node j , denoted by $d_{i,j}$. The closeness centrality of node i is defined as the inverse of the average shortest path length $d_{i,j}$, across all nodes j that i connects to:

$$v_i = \frac{n-1}{\sum_j d_{i,j}}$$

where j is any node in the same connected component with i , and n is the number of nodes in that connected component (including i). If the network is directed, then we typically focus on shortest-paths from any node j to node i (i.e., incoming paths to i). The range of closeness centrality values is limited between 0 and 1.

The closeness centrality has some shortcomings, including the fact that it does not consider all network nodes – only nodes that are in the same connected component with i . So an isolated cluster of nodes can have high closeness centrality values (close to 1) even though those nodes are not even connected to most other nodes.

Harmonic Centrality



An improved metric is harmonic centrality, defined as:

$$v_i = \sum_{j \neq i} \frac{1}{d_{i,j}}$$

Here, if nodes i and j cannot reach other, the corresponding distance can be thought of as infinite, and thus the term $\frac{1}{d_{i,j}}$ is 0. Sometimes, the harmonic centrality is normalized by $\frac{1}{n-1}$ – but that does not affect the relative ordering of node centralities.

Food For Thought

Can you think of some network analysis applications where it would make more sense to use the closeness centrality (or harmonic centrality) metric instead of the eigenvector/Katz/PageRank metrics? What about the opposite?